

Class I Tooth Preparation



Learning outcomes



1. Outlines the objectives of class I preparation
2. Describe the steps and clinical technique for class I preparation
3. Compare the differences between amalgam and composite class I preparation

OBJECTIVES OF class I TOOTH PREPARATION



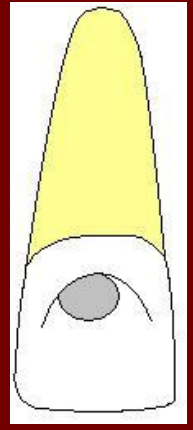
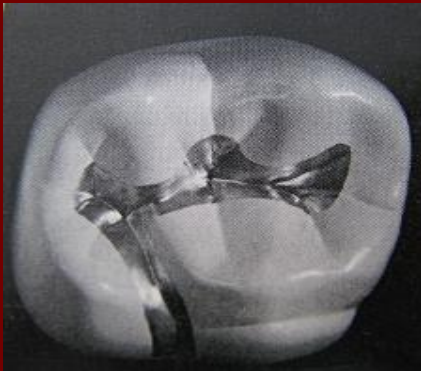
- Remove all defects .
- Provide necessary protection to the pulp.
- Form the tooth preparation, so that it will withstand masticatory forces without fracture.



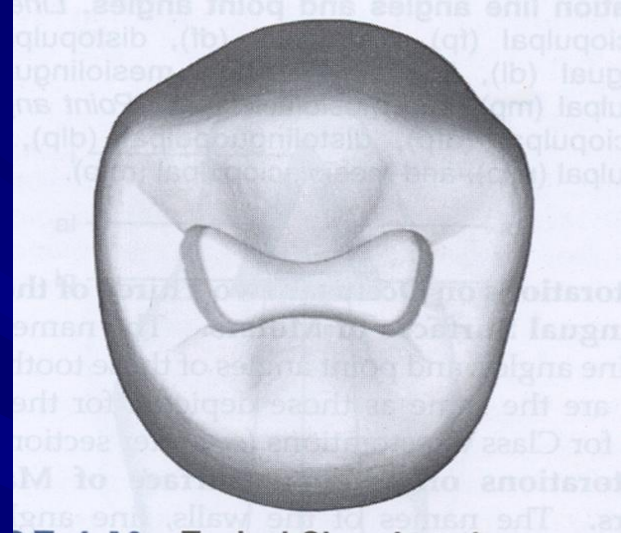
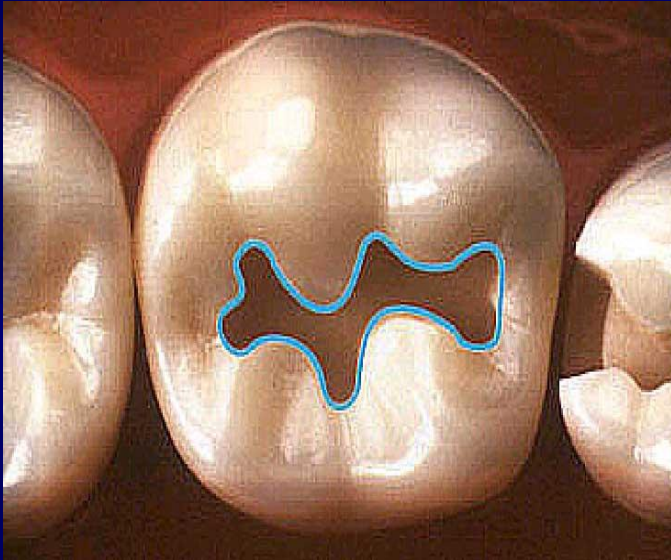
■ Class I preparation:

All pit & fissures are Class I.

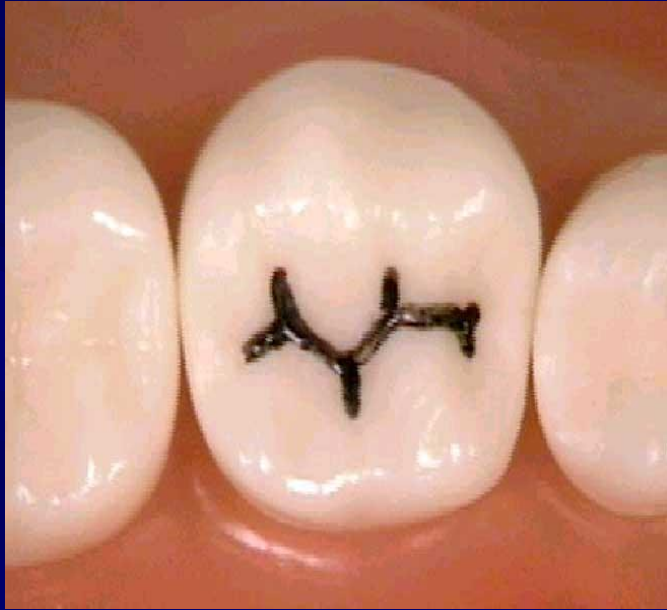
- Preparation on occlusal surface of premolars & molars.
- Preparation on occlusal two thirds of the facial & lingual surface of molars.
- Preparation on lingual surface of maxillary incisors.



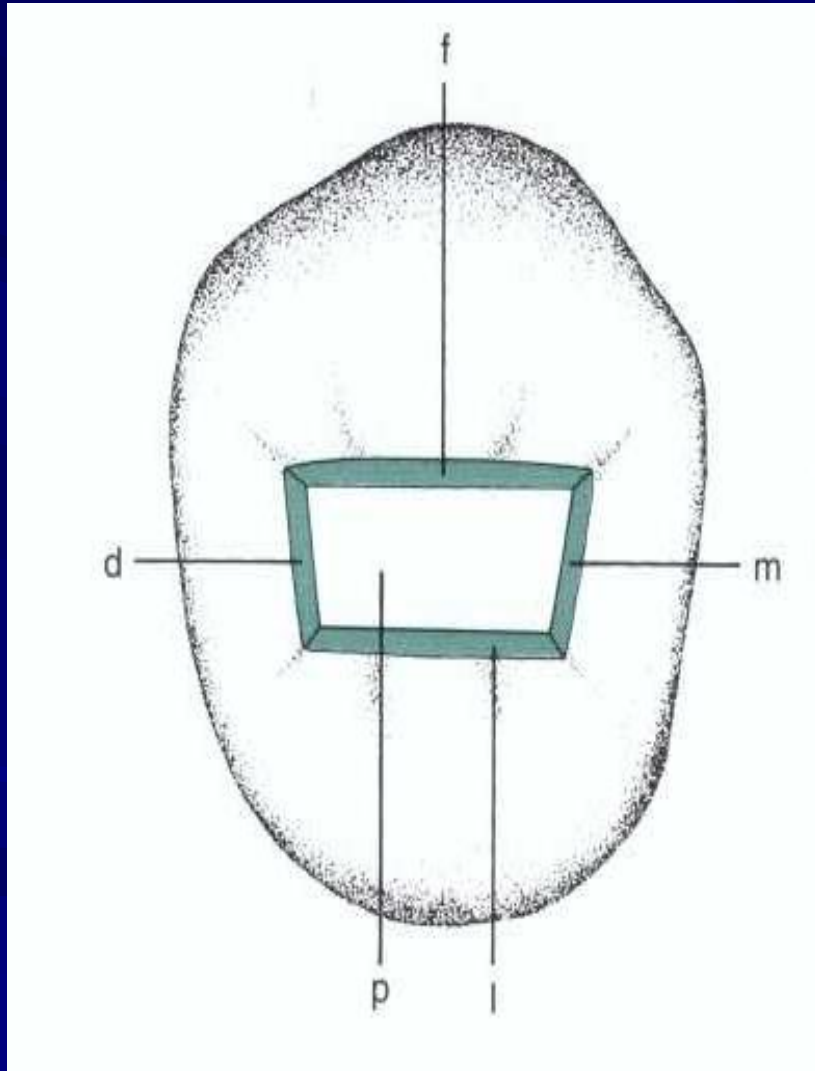
OUT LINE FORM FOR CLASS I PREPARATION



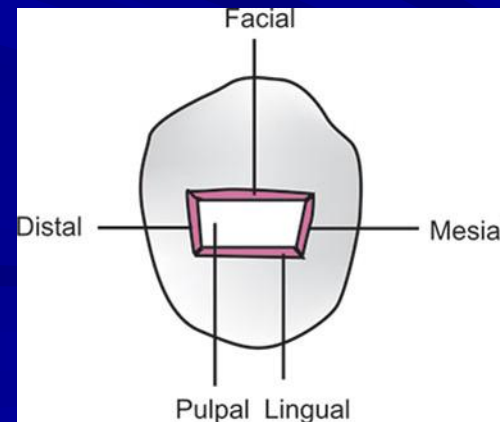
PIT AND FISSURE CARIES



CAVITY WALLS IN CLASS I PREPARATION



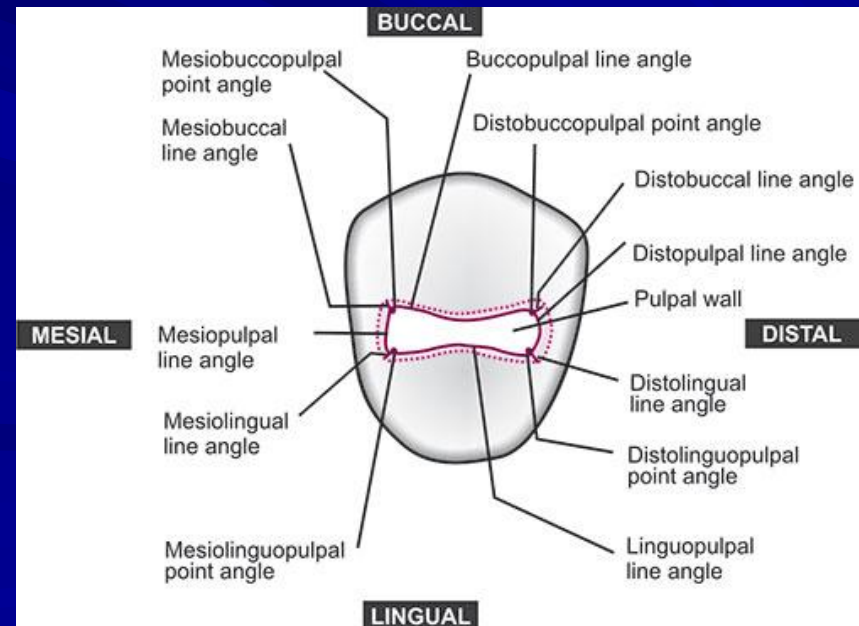
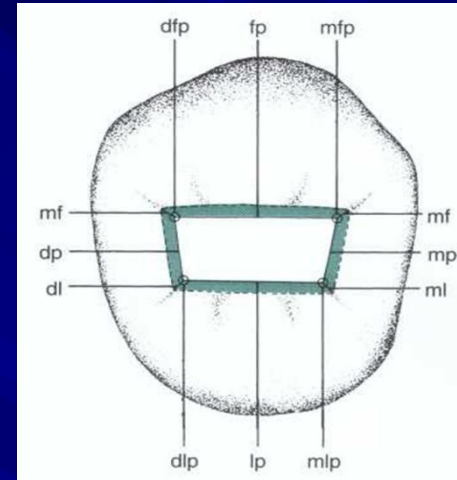
- F-Facial wall
- M-Mesial wall
- D-Distal wall
- L-Lingual wall
- P-Pulpal floor



Line Angles In Class- I :

8-Line Angles

1. Bucco-pulpal
2. Disto-buccal
3. Disto-pulpal
4. Disto-lingual
5. Linguo-pulpal
6. Mesio-lingual
7. Mesio-pulpal
8. Mesio-buccal

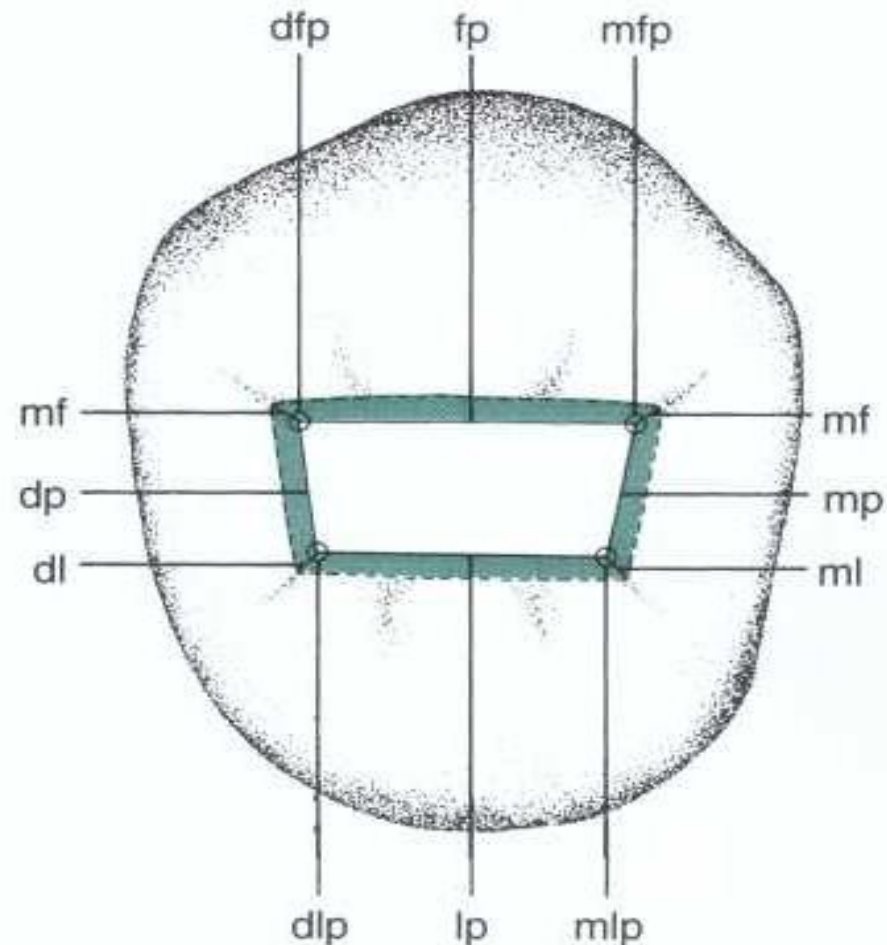


4 - Point Angles

1. Disto-facio-pulpal
2. Disto-linguo-pulpal
3. Mesio-linguo-pulpal
4. Mesio-facio-pulpal

5-Walls:

- 1.mesial wall
- 2.Distal wall
- 3.lingual/Palatal.
- 4. facial Wall.
- 5. Pulpal wall.



- Class I refers to restorations on the occlusal surfaces of posterior teeth, the occlusal two thirds of facial and lingual surfaces of posterior teeth, and the lingual surfaces of anterior teeth.
- Although most small Class I restorations are restored with a composite restoration, amalgam is still indicated in more extensive Class I restorations and tooth that cannot be well isolated



Initial Clinical Procedures



- Local anesthesia
- Shade selection, when composite restoration is planned
- Generally, isolation of the operating site with the Rubber dam is recommended.
- Tooth preparation – Initial tooth preparation
Final tooth preparation



Stages of Cavity Preparation

❑ INITIAL STEPS OF CAVITY PREPARATION



OUTLINE FORM AND INITIAL DEPTH



PRIMARY RESISTANCE FORM



PRIMARY RETENTION FORM



CONVENIENCE FORM



- The outline form for the Class I occlusal tooth preparation should include only the faulty, defective occlusal pits and fissures (in a way that sharp angles in the marginal outline are avoided).



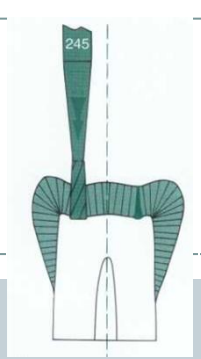
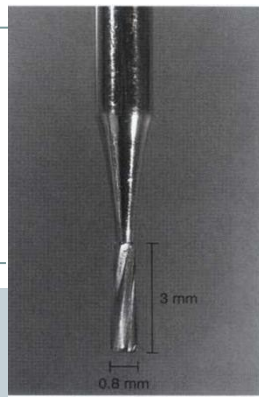
Factors governing the outline form



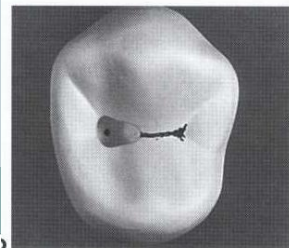
Two major factors affecting outline form

1. Carious tooth structure should be completely eliminated.
2. Margins should be placed in a sound tooth structure.





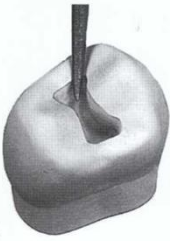
- A No. 245 bur with a head length of 3 mm and a tip diameter of 0.8mm is recommended to prepare the Class I tooth preparation.
- The Class I occlusal tooth preparation is begun by entering the deepest or most carious pit with a punch cut using the No. 245 carbide bur at high speed with air-water spray.



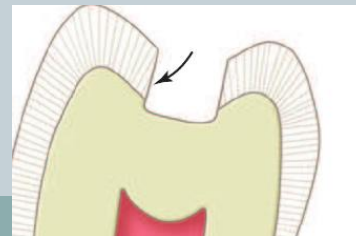


- A punch cut is performed by orienting the bur so that its long axis parallels the long axis of the tooth .
- The bur should be rotating when it is applied to the tooth and should not stop rotating until it is removed from the tooth





- One should continue to maintain the bur's orientation and depth and with intermittent pressure, extend along the central fissure toward the other pit to obtain a flat pulpal floor.
- The 1.5-mm initial pulpal depth is measured at the central fissure
- The desired pulpal depth is usually 0.1 to 0.2 mm into dentin.





- Ideally, the width of the cavity should be just wider than the diameter of the bur.
- Buccolingual width should be not more than $\frac{1}{4}$ the intercuspatal distance.

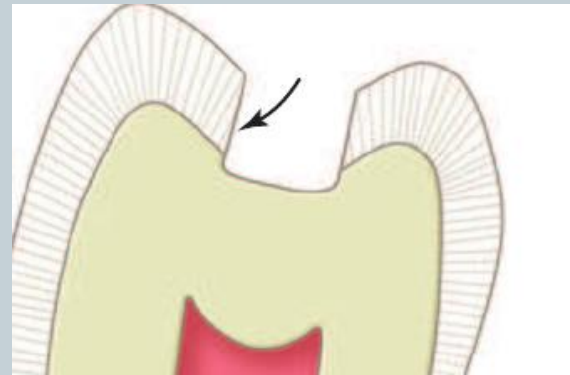
For example If intercuspatal distance is 6 mm ,then width should be

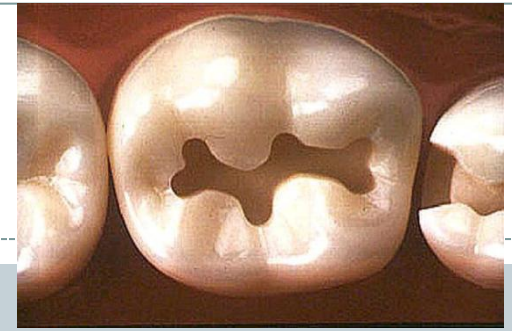
$$6/4 = 1.5 \text{ mm}$$

- It is well established that a tooth preparation with a narrow occlusal isthmus is less prone to fracture.

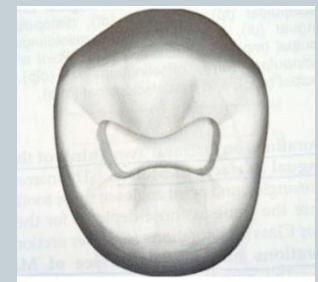


- For the conservative Class I preparation, a initial depth of 1.5 to 2 mm are considered ideal.
- The pulpal floor, depending on the enamel thickness, is usually in dentin.



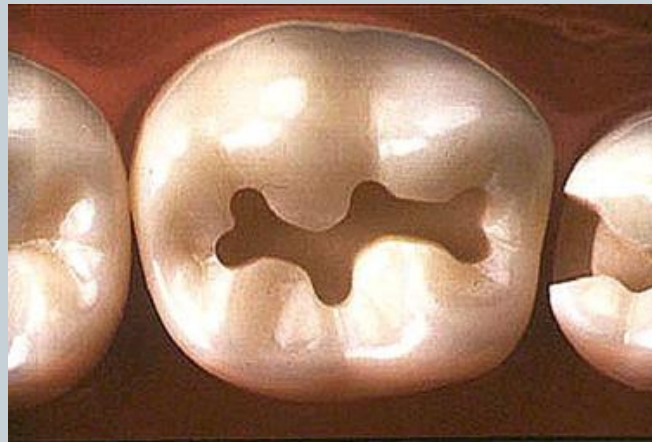


- The outline should be smoothly followed with sweeping curves
- The outline should be extended like a smooth flowing curves at the base of the cuspal slope
- The margins are placed on sound areas that are with minimal occlusal forces
- Mesial and distal walls are parallel



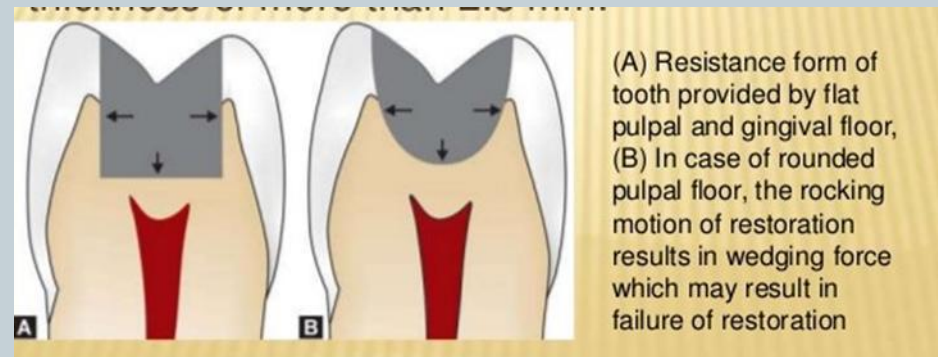
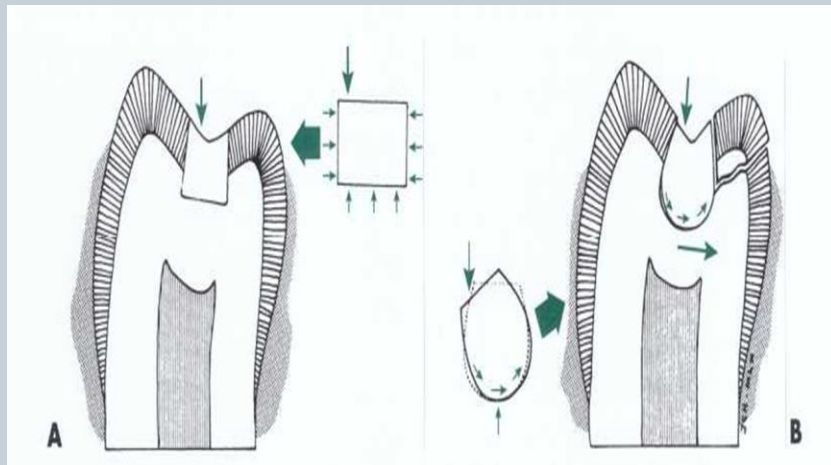
The curves should be a rounded , not angular .why?

- Because sharp angles leads to stress concentration during occlusion and it can cause fracture of tooth or restoration



PRIMARY RESISTANCE FORM is provided by the following:

- Sufficient area of relatively flat pulpal floor in sound tooth structure to resist forces directed in the long axis of the tooth and provide a strong, stable seat for the restoration.



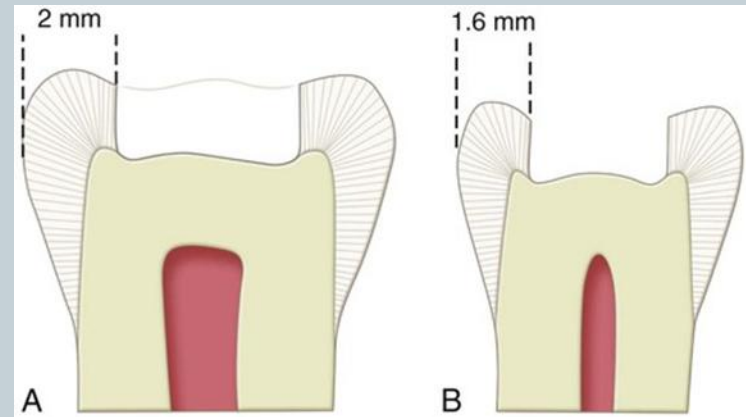


- Strong, ideal enamel margins

Marginal ridge thickness for class I tooth preparation.

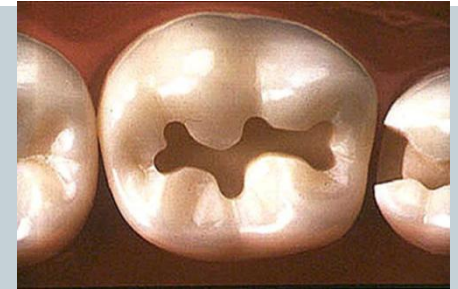
Premolar - 1.6 mm

Molar – 2 mm





- Rounded internal line angles
- Adequate depth (i.e. 1.5- 2 mm) to result in adequate thickness of the restoration, providing resistance to fracture and wear.

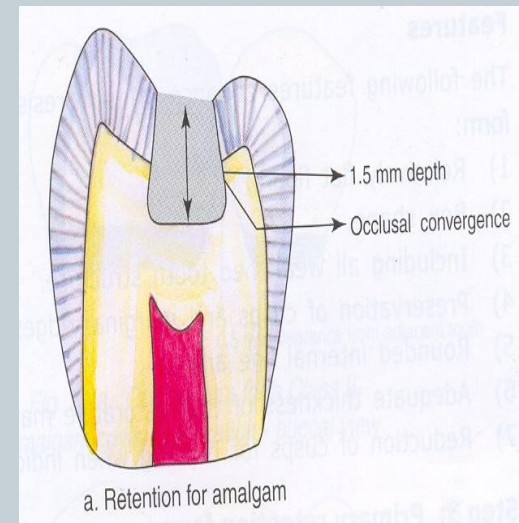
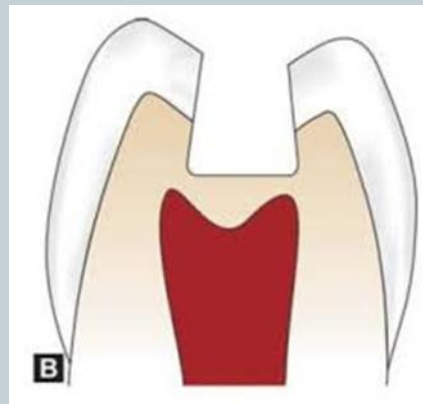
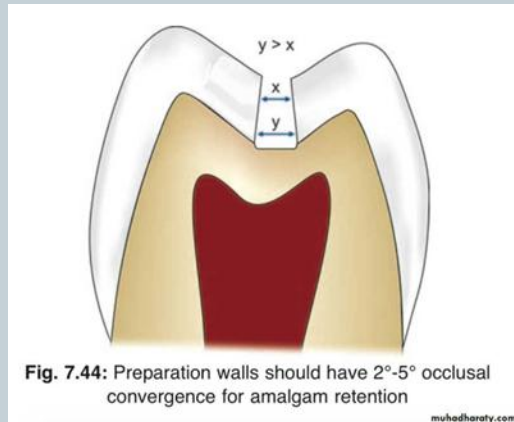


Retention form



Class I amalgam tooth preparation requires

- Occlusal convergence of facial and lingual walls .



Final Tooth Preparation

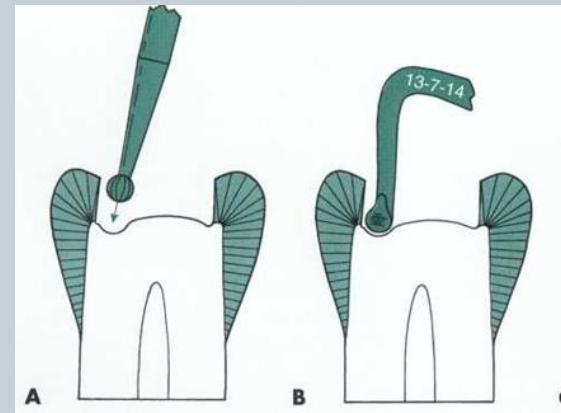


Final tooth preparation includes

- (1) Removal of remaining defective enamel and infected dentin on the pulpal floor
- (2) Pulp protection, where indicated
- (3) Procedures for finishing external walls
- (4) Final procedures of cleaning and inspecting the prepared tooth.

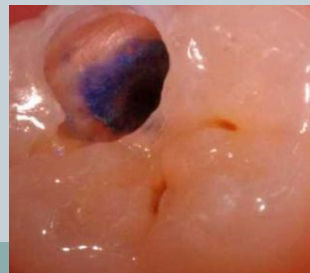


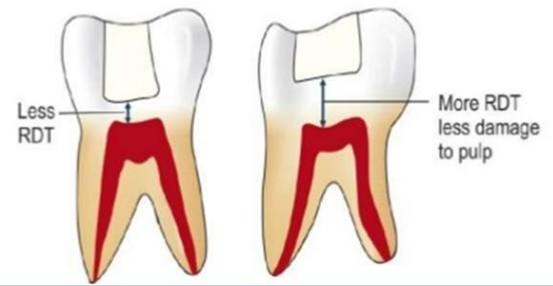
- Removal of the remaining infected dentin (i.e., caries that extends pulpally from the established pulpal floor) is best accomplished using a spoon excavator or a slowly revolving, round steel bur of appropriate size.



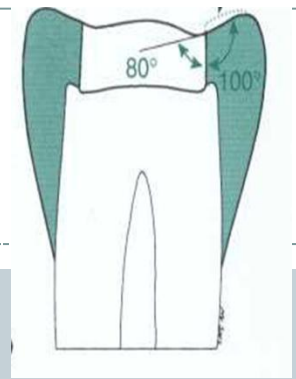


- When removing infected dentin, the excavation should be stopped when tooth structure feels hard or firm (i.e., the same feel as sound dentin).
- The dentist should ensure that caries is completely removed.
- A caries-detecting solution may be helpful in determining adequate excavation.





- If the tooth preparation is of ideal or shallow depth no liner or base is indicated.
- In deeper carious excavations (where the remaining dentin thickness- RDT (Remaining Dentin Thickness) is judged to be 0.5 - 1mm, one should place a thin layer (0.5-0.75mm) of a light-cured, resin-modified glass-ionomer (RMGIC) base
- It should be placed only over the deepest portion of the excavation.



It is important to provide an approximate 90 degree cavosurface margin for cavity preparation which should result in 80- to 90-degree amalgam at the margins

- This butt joint cavosurface margin (90 degree) of enamel and amalgam is the strongest for both.
- Amalgam is a brittle material with low edge strength and tends to chip under occlusal stress, if its angle at the margins is less than 80 degrees.

- Every completed tooth preparation should be inspected and cleaned before restoration. The tooth preparation should be free of debris after rinsing the tooth with the air-water syringe.



- A cotton pledget can also be used



Types of class I preparation for composite

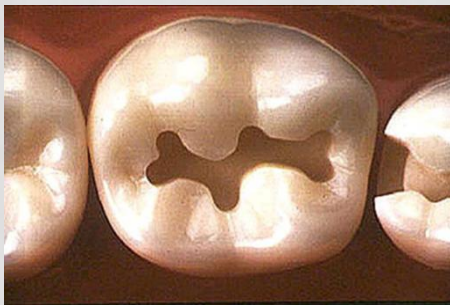


The three typical composite preparations

1. Conventional
2. Modified

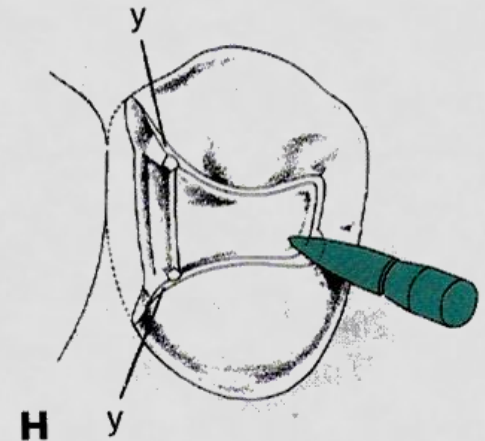
CONVENTIONAL PREPARATION FOR COMPOSITE

- Same like Amalgam type of conventional preparation form may be indicated for large preparations or restorations subjected to heavy occlusal forces.
- Mainly for large cavities or large caries.



NO OCCLUSAL BEVELING IS DONE IN CLASS I TOOTH PREPARATION

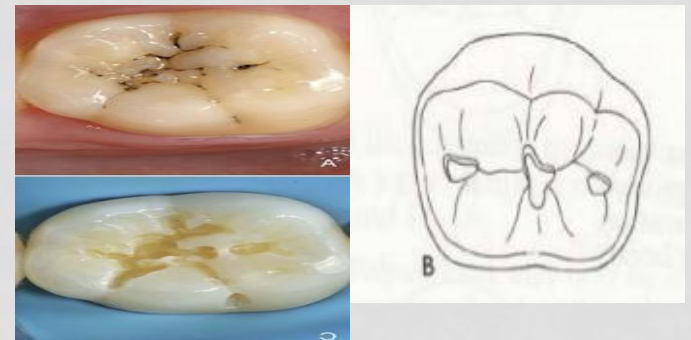
- No attempt is made to place additional beveling on the occlusal margin because it may result in thin composite in areas of heavy occlusal contact and can fracture the restoration

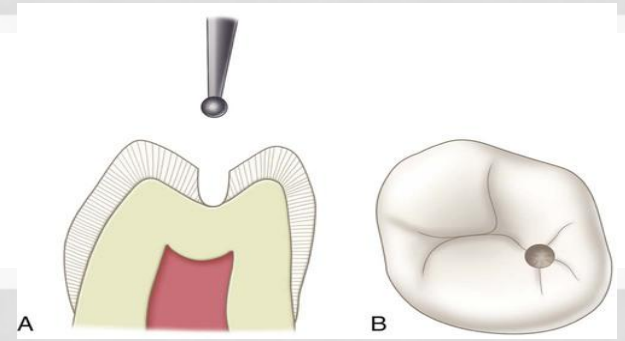


MODIFIED CLASS I TOOTH PREPARATION FOR COMPOSITE

- Minimally involved Class I lesions or faults may be restored with composite using modified tooth preparations.
- Indicated in Small to moderate Class I direct composite restorations .
- It use minimally invasive tooth preparations and do not require typical resistance and retention form features like conventional preparation.

Sturdevants art and science 5 th edition page no 578





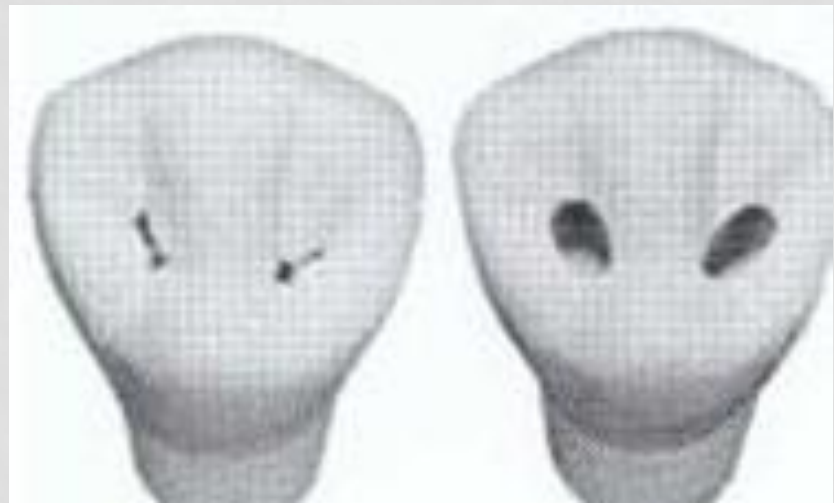
THESE PREPARATIONS ARE LESS SPECIFIC IN FORM,
HAVING A SCOOPED-OUT APPEARANCE.

THEY ARE PREPARED WITH A SMALL ROUND OR
PEAR SHAPED BUR

THE INITIAL PULPAL DEPTH IS APPROXIMATELY
0.2 MM INSIDE THE DEJ .OVERALL DEPTH 1.5 MM. BUT
MAY NOT BE UNIFORM
(THE PULPAL FLOOR IS NOT NECESSARILY TO BE FLAT
THROUGHOUT ITS LENGTH).

CLASSICAL EXAMPLE FOR MODIFIED PREPARATION DESIGN

- Mandibular premolars often have two separate faulty occlusal pits located in areas of minimal function



Amalgam vs composite Class I tooth preparation

	AMALGAM	COMPOSITE
OUTLINE FORM	Include caries or any fault	Same
Pulpal depth	Uniform 1.5 - 2 mm	Remove fault, not necessary to be uniform
Cavosurface margin	90 degree cavosurface	More than 90 degree or same like amalgam
Resistance form	Flat floor, rounded line angles, box shape	Same for larger cavity preparation No special form for smaller modified preparation
Primary Retention form	Facial and lingual walls Convergence occlusally	None, composite primary retentive form is micromechanical bond
Texture of prepared walls	Smooth	Not necessarily smooth

References



1. Sturdevants Art and science 5 th edition page no 712 -720
2. Sturdevants Art and science 6 th edition page no 576 - 578